CitationTrace

A SaaS Platform for Verified Scholarly Citation and Provenance Preservation

SC4052 Cloud Computing Project

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Abstract

The rapid integration of large language models (LLMs) into scholarly workflows has introduced a systemic breakdown in the chain of academic credit , a crisis recently formalized as the provenance problem. LLMs routinely reproduce intellectual insights without citing their origins, and empirical benchmarks show that leading systems fabricate citations 39-90% of the time. This report presents CitationTrace, a SaaS platform that addresses the provenance problem through a retrieval-augmented generation (RAG) pipeline with integrated citation verification, confidence scoring, and end-to-end audit logging. CitationTrace retrieves real papers from scholarly databases, generates synthesized answers with inline citations, and verifies every claimed reference against retrieved abstracts using semantic similarity. The platform provides an auditable, reproducible chain of evidence from every claim to its source, making attribution automatic and mandatory rather than relying on researcher diligence. We situate this work within the broader literature on AI-generated hallucinations, citation faithfulness, and trustworthy RAG systems, demonstrating how CitationTrace bridges ethical gaps in current AI-assisted research.

Keywords: citation verification, provenance preservation, retrieval-augmented generation, scholarly attribution, LLM hallucination, trustworthy AI

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1. Introduction

The emergence of large language models as tools for scholarly research has fundamentally altered how knowledge is produced, consumed, and attributed. Systems such as GPT-4o, GPT-5, Gemini, Perplexity, and Copilot are now routinely used by researchers to draft literature reviews, summarize findings, and generate citations. Yet these same systems suffer from a deeply consequential flaw: they fabricate scholarly references at alarming rates, producing citations that appear legitimate but point to nonexistent papers, incorrect authors, or fabricated DOIs.

Earp et al. (2025) recently formalized this crisis as the provenance problem , a systematic breakdown in the chain of scholarly credit wherein AI reproduces intellectual insights without citing their origins, even when users disclose AI use and act in good faith. The problem is distinct from traditional plagiarism because it is non-intentional: the researcher may have no way of knowing that the AI has failed to attribute a claim to its source. Current ethical and professional frameworks, which treat attribution failure through a plagiarism lens focused on intentional deception, are inadequate for this new category of provenance failure.

The empirical evidence is stark. The OpenScholar benchmarks published in Nature in February 2026 demonstrated that GPT-4o fabricates citations 78-90% of the time, while GPT-5 , despite significant improvements , still exhibits a title-level hallucination rate of 39% (Asai et al., 2026). A GPTZero analysis found that at least 51 papers accepted at NeurIPS 2025 contained AI-hallucinated citations (Gibney, 2026). The DeepTRACE framework (Venkit et al., 2025) confirmed that citation accuracy across major systems ranges from only 40-80%, with deep-research configurations still producing large fractions of unsupported statements.

These findings motivate the present work. CitationTrace is a SaaS platform designed to address the provenance problem at its root by making citation verification automatic, mandatory, and auditable. Rather than relying on researchers to manually check AI-generated references , a task that grows intractable as AI-scale knowledge production accelerates , CitationTrace embeds verification directly into the generation pipeline. Every claim is traced to a real, retrievable source; every citation is verified against its abstract using semantic similarity; and every step is logged in an auditable chain of evidence.

This report proceeds as follows. Section 2 reviews the literature on citation hallucination, the provenance problem, retrieval-augmented generation (RAG), and trustworthy AI. Section 3 formally states the problem. Section 4 describes the CitationTrace platform architecture and design. Section 5 presents illustrative examples. Section 6 offers conclusions and directions for future work.

2. Literature Review

This section surveys the key bodies of work that inform CitationTrace: the provenance problem and its ethical dimensions, empirical evidence of citation hallucination, advances in retrieval-augmented generation for citation faithfulness, frameworks for trustworthy AI, and early work on autonomous and traceable AI research systems.

2.1 The Provenance Problem

Earp et al. (2025), in their paper "The Provenance Problem: LLMs and the Breakdown of Citation Norms" (arXiv:2509.13365), provide the most comprehensive articulation of the attribution crisis introduced by LLMs. They define the provenance problem as a systematic breakdown in the chain of scholarly credit: AI systems reproduce intellectual insights , sometimes paraphrased, sometimes nearly verbatim , without citing the original sources from which those insights were derived during training.

Crucially, Earp et al. argue that current ethical and professional frameworks fail for three reasons:

> First, existing frameworks treat attribution failure through a plagiarism lens, which presupposes intentional deception. But provenance failure in AI-assisted writing is non-intentional , the researcher may genuinely not know that the AI has failed to cite a source, and no amount of good-faith diligence can fully resolve the gap.
>
> Second, publisher policies addressing AI use are fragmented and insufficient for AI-scale knowledge production. Different journals and conferences impose different requirements, and none provides a mechanism for systematically verifying that AI-generated content properly attributes its sources.
>
> Third, no governance mechanism currently exists to trace AI training data back to original scholarly sources. The training corpora of frontier LLMs are opaque, and there is no standardized way to determine which papers informed a model's outputs.

This analysis establishes the ethical foundation for CitationTrace: a system that addresses non-intentional provenance harm by making attribution automatic and mandatory, rather than relying on researcher diligence within a broken framework.

2.2 Citation Hallucination: Empirical Evidence

A growing body of empirical work quantifies the severity of citation hallucination across LLM systems. We review the three most significant contributions.

OpenScholar and the Nature 2026 Benchmarks.

Asai et al. (2026), published in Nature (DOI:), introduced OpenScholar, an open-source retrieval-augmented language model (RALM) designed specifically for scientific question answering. The paper established ScholarQABench, a benchmark of 2,967 queries, and introduced the OpenScholar DataStore (OSDS) ,a database of 45 million papers. OpenScholar, available in 8B and 70B parameter variants, employs a pipeline consisting of a retriever, reranker, self-feedback loop, and post-hoc attribution mechanism.

The benchmarks revealed devastating findings about frontier LLMs:

- GPT-4o fabricates citations 78-90% of the time across evaluated domains.
- GPT-5, while improved, still exhibits title-level hallucination at 39%, with fabricated citations remaining common.
- OpenScholar outperformed GPT-4o by 6.1% and PaperQA2 by 5.5% on ScholarQABench, achieving citation F1 scores of 47-64% and expert-level citation accuracy.
- A GPTZero analysis identified at least 51 NeurIPS 2025 papers containing AI-hallucinated citations.

Gibney (2026), writing in Nature (DO), reported on these findings, highlighting that OpenScholar's open-source approach demonstrated that smaller, purpose-built models with proper retrieval infrastructure can outperform giant commercial LLMs for scholarly tasks.

DeepTRACE.
Venkit et al. (2025) introduced DeepTRACE (arXiv:2509.04499), a comprehensive framework for evaluating LLM-generated research responses across eight measurable dimensions spanning answer text, sources, and citations. DeepTRACE employs statement-level analysis , decomposing responses into individual claims, assigning confidence scores, and constructing citation matrices and factual-support matrices.

Applying this framework to GPT-4.5, GPT-5, Perplexity, Copilot/Bing, and Gemini, Venkit et al. found that citation accuracy ranges from 40% to 80% across systems. Even deep-research configurations , modes specifically designed for thorough, cited research , still exhibit large fractions of unsupported statements. This finding is particularly significant because it demonstrates that the problem persists even when systems are explicitly optimized for citation accuracy.

Early Evidence of Fabrication.
Emsley (2023) provided early evidence of this problem in a study published in Schizophrenia (DOI:), documenting that ChatGPT routinely fabricates plausible-looking but entirely nonexistent citations when asked to produce academic references. This early warning was subsequently confirmed at scale by the OpenScholar and DeepTRACE evaluations.

2.3 Retrieval-Augmented Generation and Citation Faithfulness

Retrieval-augmented generation (RAG) , the practice of grounding LLM outputs in retrieved documents , has emerged as the primary technical strategy for reducing hallucination and improving citation faithfulness. Several recent systems advance the state of the art.

HySemRAG. Godinez (2025) introduced HySemRAG (arXiv:2508.05666), a hybrid semantic retrieval and agentic quality assurance system that achieves 99% citation accuracy. HySemRAG combines dense retrieval with semantic verification and an agentic loop that iteratively checks and corrects citations before presenting results to users. This represents a near-ceiling performance on citation faithfulness, demonstrating that the technical problem is solvable given sufficient architectural investment.

ScholarCopilot. Wang et al. (2025) developed ScholarCopilot (arXiv:2504.00824), a RAG system specifically designed for academic writing that integrates retrieval directly into the generation process. Unlike post-hoc citation attachment, ScholarCopilot interleaves retrieval and generation, producing text with accurate inline citations as a natural part of the writing process rather than as an afterthought.

CiteCheck. Xu et al. (2025) introduced CiteCheck (arXiv:2502.10881), a dedicated tool for citation faithfulness detection. CiteCheck evaluates whether the claims made in a text are actually supported by the cited sources, providing a verification layer that can be applied to any AI-generated content.

CoV-RAG. He et al. (2024) proposed Chain-of-Verification RAG (CoV-RAG, arXiv:2410.05801), which applies a chain-of-verification approach to RAG outputs. Rather than accepting the initial retrieval and generation at face value, CoV-RAG decomposes claims, verifies each against the retrieved sources, and iteratively refines the output ,significantly reducing hallucination rates.

2.4 Trust, Transparency, and Trustworthy RAG

The broader question of trustworthiness in RAG systems has been systematically addressed by Zhou et al. (2024) in their comprehensive survey (arXiv:2409.10102). They identify six critical dimensions of trustworthy RAG: factuality (are the outputs accurate?), robustness (do they degrade gracefully under adversarial or noisy inputs?), fairness (are biases propagated from retrieval?), transparency (can users understand how answers were derived?), accountability (can the system's decisions be traced and audited?), and privacy (are user queries and source documents protected?).

This six-dimensional framework provides the theoretical grounding for CitationTrace's design. Our platform directly addresses factuality (through citation verification), transparency (through the chain-of-evidence audit log), and accountability (through LangSmith end-to-end tracing).

Ding et al. (2025) contributed an important finding regarding the psychology of AI-generated citations (arXiv:2501.01303). Their study revealed that trust in LLM responses increases even when citations are random and unsupported , but, critically, trust decreases when users actually check the citations and find them to be inaccurate. This asymmetry underscores the danger: citations create a veneer of credibility that most users never pierce, but those who do verify find the trust fundamentally violated. CitationTrace addresses this by performing verification proactively, before the user ever encounters the citation.

Huang et al. (2024) provided a comprehensive survey of LLM hallucination (arXiv:2311.05232), developing a taxonomy of hallucination types and introducing the Hallucination Vulnerability

Index (HVI). Their work categorizes hallucination into intrinsic (contradicting the source) and extrinsic (unsupported by the source) varieties, and identifies citation hallucination as a particularly consequential subtype because it undermines the epistemic infrastructure of science.

2.5 Toward Autonomous and Traceable AI Research

Ifargan et al. (2024) examined the requirements for autonomous LLM-driven research (arXiv:2404.17605), identifying three pillars that any responsible AI research system must satisfy: traceability (every output can be traced to its inputs and reasoning steps), transparency (the system's decision-making process is interpretable), and verifiability (claims can be independently checked against their sources).

García-Barriocanal et al. (2021) established foundational principles for traceability in trustworthy AI (DOI:), arguing that traceability is not merely a technical convenience but a prerequisite for AI systems that operate in high-stakes domains such as academic research. Their framework emphasizes that traceability must be end-to-end , from data ingestion through processing to output , and must be designed into the system architecture rather than bolted on as an afterthought.

Together, these works establish the theoretical and empirical foundations upon which CitationTrace is built: the provenance problem demands a solution; existing systems fail to provide one; RAG with integrated verification represents the most promising technical approach; and the resulting system must be transparent, traceable, and verifiable by design.

3. Problem Statement

The central problem addressed by CitationTrace can be stated as follows:

> Large language models used in scholarly workflows routinely generate citations that are fabricated, inaccurate, or unsupported by the cited source , at rates ranging from 39% to 90% depending on the system and configuration. Current ethical frameworks, publisher policies, and technical tools are insufficient to address this provenance failure because they either assume intentional misconduct, operate in fragmented and inconsistent ways, or lack mechanisms to trace AI-generated claims back to real scholarly sources. There is a need for an integrated, auditable platform that makes citation verification automatic, mandatory, and transparent.

This problem decomposes into several sub-problems that CitationTrace must address:

1. Citation Fabrication. LLMs generate references to papers that do not exist. GPT-4o fabricates citations 78-90% of the time (Asai et al., 2026). Even improved models like GPT-5 hallucinate titles at a 39% rate. The system must ensure that every cited paper actually exists and is retrievable.

2. Citation Misattribution. Even when a cited paper exists, the claims attributed to it may not be supported by the paper's actual content. DeepTRACE found that citation accuracy ranges 40-80% (Venkit et al., 2025). The system must verify semantic alignment between claims and cited sources.

3. Non-Intentional Provenance Failure. Researchers using AI in good faith cannot detect provenance failures without manual verification of every citation , a task that is impractical at scale. The system must automate this verification without requiring user intervention.

4. Absence of Audit Trails. Current AI systems provide no mechanism for third parties to inspect how an answer was derived, which sources were consulted, or why a particular citation was selected. The system must provide a complete, reproducible chain of evidence.

5. Trust Asymmetry. Users trust AI-generated citations without checking them (Ding et al., 2025), creating a false sense of credibility. The system must surface verification status proactively, making the reliability of each citation immediately visible.

6. Fragmented Governance. Publisher policies on AI use are inconsistent and provide no unified verification mechanism. The system must offer a standardized, platform-level approach to citation provenance that does not depend on individual publishers or journals.

CitationTrace is designed to address all six sub-problems simultaneously through an integrated architecture that combines retrieval-augmented generation, semantic citation verification, confidence scoring, and end-to-end audit logging.

4. Solution Design

CitationTrace is a SaaS platform composed of a FastAPI backend implementing a RAG pipeline with integrated citation verification, and a React frontend that presents results with inline citations, confidence scores, and an auditable chain of evidence. This section describes each component in detail.

4.1 Architecture Overview

The platform follows a three-tier architecture. The retrieval tier queries scholarly databases (Semantic Scholar API, CrossRef) to obtain real, existing papers relevant to the user's research question. The generation tier uses an LLM to synthesize a response grounded in the retrieved papers, producing inline citations. The verification tier evaluates every generated citation against the retrieved abstracts using semantic similarity, producing confidence scores and verification verdicts. All three tiers are instrumented with LangSmith for end-to-end tracing.

Tier	Component	Technology	Function
Retrieval	Paper Fetcher	Semantic Scholar API / CrossRef	Retrieve real papers matching query terms
Retrieval	Reranker	sentence-transformers	Rank papers by relevance to query
Generation	Synthesizer	LLM (GPT-4o open-source)	Generate answer with inline [n] citations
Verification	Citation Verifier	sentence-transformers (cosine sim.)	Verify each citation abstract
Verification	Confidence Scorer	Composite scoring module	Assign 0–1 confidence statement
Audit	Trace Logger	LangSmith	End-to-end logging of every LLM call

Table 1. CitationTrace architecture components by tier.

4.2 Backend: FastAPI + RAG Pipeline

The backend is implemented in Python using FastAPI, chosen for its asynchronous request handling, automatic OpenAPI documentation, and ease of deployment. The primary endpoint is /query, which accepts a research question and returns a JSON response containing the synthesized answer, cited sources, confidence scores, and the full audit trail.

Retrieval Pipeline. Upon receiving a query, the system:

1. Extracts key terms and concepts from the research question using an LLM-based query decomposition step.

2. Queries the Semantic Scholar API (with CrossRef as a fallback) to retrieve candidate papers , including titles, authors, abstracts, DOIs, and publication years.

3. Reranks the retrieved papers using a sentence-transformer model that computes semantic similarity between the query and each paper's abstract, selecting the top-k most relevant papers (default k = 10). 4. Constructs a grounding context from the selected papers' abstracts and metadata, which is provided to the LLM as the sole basis for generating its response.

Generation. The LLM receives the grounding context along with explicit instructions to:

- Synthesize a comprehensive answer to the research question.
- Cite sources using inline [n] markers corresponding to the retrieved papers.
- Only make claims that are directly supported by the provided abstracts.
- Flag any claims for which the available evidence is ambiguous or insufficient.

The generation step is instrumented with LangSmith, which records the complete prompt, the model's response, token counts, latency, and any intermediate reasoning steps. This produces a fully reproducible record of how each answer was generated.

4.3 Citation Verifier Module

The citation verifier is the core innovation of CitationTrace. After the LLM generates a response, the verifier performs a multi-step analysis of every claimed citation:

Step 1: Statement Decomposition.

Following the approach pioneered by DeepTRACE (Venkit et al., 2025), the generated response is decomposed into individual statements, each paired with its claimed citation(s). This mirrors DeepTRACE's statement-level analysis, which builds citation matrices mapping each claim to its purported sources.

Step 2: Semantic Similarity Verification.

For each statement-citation pair, the verifier computes the cosine similarity between the statement's embedding and the cited paper's abstract embedding, using a pre-trained sentence-transformer model (e.g., all-MiniLM-L6-v2). This produces a similarity score in the range [0, 1].

Step 3: Confidence Scoring.

Each statement receives a composite confidence score that integrates:

- Semantic similarity between the claim and the cited abstract (primary signal).
- Retrieval rank of the cited paper (higher-ranked papers receive a small bonus).
- Cross-reference consistency (if multiple papers support the same claim, confidence increases). Step 4: Verification Verdict.

Based on the composite confidence score, each citation receives a verification status:

Status	Confidence Range	Meaning
Verified	0.70 - 1.00	Claim is semantically supported by the cited abstract
Partially Verified	0.40 - 0.69	Some support found; claim may extend beyond the abstract
Unverified	0.00 - 0.39	No meaningful semantic alignment between claim and cited source

Table 2. Citation verification status thresholds.

Step 5: Chain-of-Evidence Audit Log.

Every verification step is recorded in a structured audit log: claim text → cited source (title, DOI, abstract snippet) →embedding similarity score → composite confidence score → verification status. This log is stored alongside the LangSmith trace, producing a complete, reproducible record that third parties can inspect. The approach aligns with the traceability, transparency, and verifiability pillars identified by Ifargan et al. (2024) and the traceability-for-trustworthy-AI framework of García-Barriocanal et al. (2021).

4.4 Frontend: React Application

The frontend is a React single-page application designed around three primary panels:

Query Input Panel.

A clean input interface where the user enters a research question. The interface supports both natural language queries ("What are the effects of RAG on citation hallucination rates?") and structured queries with domain constraints.

Response Panel.

Displays the synthesized answer text with inline [1], [2], [3] citation markers. Each marker is clickable and scrolls to the corresponding entry in the Citation Panel. The response text is formatted for readability with clear paragraph breaks and highlighted key findings.

Citation Panel.

Each cited source is displayed as a card containing: the paper title, author list, DOI (as a clickable link), an abstract snippet, the computed confidence score (0-1), and the verification status displayed as a prominent badge. Verified citations display a green checkmark badge; unverified citations display an orange warning indicator. This design directly addresses the trust asymmetry identified by Ding et al. (2025) , rather than allowing users to assume citations are valid, the system surfaces verification status proactively.

Audit Trail Panel.

An expandable panel that displays the full chain-of-evidence log for each claim. Users can inspect the retrieval step (which papers were fetched and why), the generation step (what prompt was sent to the LLM), and the verification step (how each citation was scored). For deeper inspection, the panel links to the full LangSmith trace, which records every LLM call, retrieval query, and intermediate computation.

4.5 Bridging Ethical Gaps

CitationTrace is designed to address each of the three ethical gaps identified by Earp et al. (2025):

Ethical Gap	How CitationTrace Addresses It
Non-intentional provenance harm Existing framework treats attribution failure as intentional deception.	CitationTrace makes attribution automatic and mandatory. Every generated claim is linked to a real source through the RAG pipeline, and every link is verified before presentation. The researcher need not rely on diligence alone , the system enforces provenance by design.
Fragmented governance Publisher policies are inconsistent and insufficient for AI-scale production.	CitationTrace provides an auditable, reproducible trace (via LangSmith) that any third party , journal editor, reviewer, institution , can inspect. This creates a standardized provenance record independent of publisher-specific policies.
Training data opacity No mechanism to trace AI outputs to original scholarly sources.	While CitationTrace cannot retroactively trace training data, it sidesteps the problem entirely by grounding every claim in real-time retrieval from scholarly databases. Claims are traced to retrieved papers, not to opaque training corpora. This follows the OpenScholar approach (Asai et al., 2026) of using a curated, inspectable knowledge base (45M papers).

Table 3. How CitationTrace bridges the ethical gaps identified in the provenance problem.

Beyond these three primary gaps, CitationTrace also advances epistemic justice. Following the OpenScholar model, the platform is designed to be open-source and low-cost, ensuring that researchers at under-resourced institutions have access to the same citation-verification infrastructure as those at well-funded universities. This accessibility dimension is critical: if citation verification is available only to privileged researchers, the provenance problem will disproportionately harm scholars who lack the resources to manually check every AI-generated reference.

5. Illustrative Examples

This section presents a detailed walkthrough of CitationTrace in operation, demonstrating how the platform handles a representative research query from submission through verification.

5.1 Example Query

Query: "What are the effects of retrieval-augmented generation on citation hallucination rates?"

5.2 Retrieval Step

The system decomposes the query into key concepts: retrieval-augmented generation, citation hallucination, and hallucination rates. It queries the Semantic Scholar API and retrieves the following five papers (ranked by relevance):

#	Paper	Relevance
[1]	Asai et al. (2026). OpenScholar: Synthesizing Scientific Literature with RALMs. Nature.	0.94
[2]	Venkit et al. (2025). DeepTRACE: Evaluating LLM-Generated Research Responses.	0.91
[3]	He et al. (2024). CoV-RAG: Chain-of-Verification RAG.	0.87
[4]	Huang et al. (2024). A Survey on Hallucination in LLMs.	0.83
[5]	Godinez (2025). HySemRAG: Hybrid Semantic RAG with Agentic QA.	0.79

Table 4. Retrieved papers ranked by relevance to the example query.

5.3 Generation Step

Using the retrieved abstracts as grounding context, the LLM generates the following synthesized response:

> "Retrieval-augmented generation (RAG) has emerged as the primary technical approach to reducing citation hallucination in LLM outputs. The OpenScholar benchmarks demonstrated that standard LLMs like GPT-4o fabricate citations 78-90% of the time, while RAG-based systems such as OpenScholar achieve citation F1 scores of 47-64% [1]. Statement-level evaluation using the DeepTRACE framework confirmed that citation accuracy across major systems ranges from 40-80%, with deep-research configurations still exhibiting significant unsupported claims [2]. Chain-of-verification approaches like CoV-RAG further reduce hallucination by iteratively verifying claims against retrieved sources [3]. A comprehensive survey of LLM hallucination identified citation fabrication as a particularly consequential subtype that undermines the epistemic infrastructure of science [4]. Most recently, HySemRAG demonstrated that hybrid semantic retrieval combined with agentic quality assurance can achieve 99% citation accuracy, suggesting that the technical problem is solvable with sufficient architectural investment [5]."

5.4 Verification Step

The citation verifier decomposes the response into individual claim-citation pairs and computes semantic similarity against the cited abstracts. Results:

Citation	Claim Summary	Cosine Sim.	Confidence	Status
[1] Asai et al.	GPT-4o fabricates OpenScholar F1 47-64%	0.89	0.91	Verified
[2] Venkit et al.	Citation accuracy across systems	0.85	0.87	Verified
[3] He et al.	CoV-RAG reduces hallucination via verification	0.78	0.81	Verified
[4] Huang et al.	Citation fabrication undermines science	0.72	0.75	Verified
[5] Godinez	HySemRAG achieves citation accuracy	0.38	0.42	Low Confidence

Table 5. Citation verification results for the example query.

In this example, citations [1] through [4] are verified with high confidence , the claims made in the generated text are semantically supported by the respective paper abstracts. Citation [5], however, receives a low confidence score of 0.42. Investigation reveals that while HySemRAG does claim 99% citation accuracy, the specific phrasing in the generated text ('hybrid semantic retrieval combined with agentic quality assurance') extends slightly beyond what the retrieved abstract explicitly states. The system flags this citation with an orange warning indicator, alerting the user to verify it manually.

5.5 Audit Trail

The full audit trail for this query is available in both the Audit Trail Panel and LangSmith. A simplified excerpt of the chain-of-evidence log:

```
Claim 1: 'GPT-4o fabricates citations 78-90% of the time' → Source: Asai et al. (2026), Nature, DOI:10.1038/s41586-025-10072-4 → Abstract match: 'GPT-4o...produces fabricated ' citations 78-90%' → Cosine similarity: 0.89 → Confidence: 0.91 → Status: Verified
```

```
Claim 5: 'HySemRAG...99% citation accuracy' → Source: Godinez (2025), arXiv:2508.05666 → Abstract match: 'achieves...99% citation accuracy' → Cosine similarity: 0.38 → Confidence: 0.42 → Status: Low Confidence (phrasing extends beyond abstract)
```

This audit trail enables any third party , journal editor, reviewer, or institution , to independently verify how each claim in the generated response was sourced, scored, and verified. The full LangSmith trace provides additional depth, including the exact prompts sent to the LLM, the raw retrieval results, and intermediate reasoning steps.

5.6 Comparison with Unverified LLM Output

To illustrate the value of CitationTrace's verification layer, consider what happens when the same query is submitted to a standard LLM without retrieval augmentation or verification. A typical GPT-4o response might include citations such as:

- "Smith, J., & Lee, K. (2024). Hallucination Reduction in RAG Systems. Journal of AI Research, 45(2), 112-129.", This paper does not exist.
- "Chen, W. et al. (2023). Citation Accuracy in Large Language Models. NeurIPS 2023." , Fabricated: no such paper in NeurIPS 2023 proceedings.

As documented by Asai et al. (2026), GPT-4o fabricates citations 78-90% of the time. Without CitationTrace's verification layer, these fabricated references would appear legitimate to the user , and, as Ding et al. (2025) showed, the mere presence of citations increases user trust even when the citations are random and unsupported.

6. Conclusions and Future Work

6.1 Summary of Contributions

This report has presented CitationTrace, a SaaS platform that addresses the provenance problem in AI-assisted scholarly research. Our contributions are threefold:

1. Architectural integration of verification. CitationTrace embeds citation verification directly into the generation pipeline, rather than treating it as a post-hoc step. Every claim is grounded in real-time retrieval from scholarly databases, verified against source abstracts using semantic similarity, and scored for confidence , before the user ever sees the output.
2. End-to-end auditability. Through LangSmith integration and the chain-of-evidence audit log,

CitationTrace produces a complete, reproducible record of how every answer was generated, which sources were consulted, and how each citation was verified. This addresses the accountability dimension identified by Zhou et al. (2024) and the traceability requirements articulated by Ifargan et al. (2024) and
García-Barriocanal et al. (2021).

3. Ethical alignment. CitationTrace directly addresses the three ethical gaps identified by Earp et al. (2025) ,non-intentional provenance harm, fragmented governance, and training data opacity , by making attribution automatic, providing standardized audit trails, and grounding claims in inspectable retrieval rather than opaque training corpora.

6.2 Limitations

Several limitations should be acknowledged. First, semantic similarity between a claim and an abstract is a necessary but not sufficient condition for citation accuracy. A claim may be semantically similar to an abstract while still misrepresenting the paper's findings , a subtlety that requires deeper content analysis beyond what cosine similarity captures. Future work should explore entailment-based verification (as in CiteCheck by Xu et al., 2025) to strengthen this layer.

Second, CitationTrace currently verifies citations against paper abstracts rather than full texts. While abstracts are universally available and provide a strong signal, full-text verification would increase accuracy for claims based on specific methods, results, or supplementary data. Integration with open-access full-text repositories is a natural extension.

Third, the system's retrieval quality is bounded by the coverage of its upstream databases (Semantic Scholar, CrossRef). Papers not indexed in these databases cannot be retrieved or cited, potentially introducing a coverage bias. Expanding the retrieval infrastructure to include preprint servers, institutional repositories, and domain-specific databases would mitigate this limitation.

Fourth, the current confidence scoring model is heuristic-based. A learned scoring model, trained on human judgments of citation appropriateness, would likely yield more calibrated confidence estimates.

6.3 Future Directions

Several promising directions emerge from this work:

- Full-text verification. Extending the citation verifier to analyze full paper texts using chunked retrieval and entailment models, following the ScholarCopilot approach of interleaving retrieval and generation at the paragraph level (Wang et al., 2025).
- Multi-hop claim verification. Implementing chain-of-verification strategies (He et al., 2024) to handle claims that require synthesis across multiple sources.
- Community-scale deployment. Deploying CitationTrace as a publicly accessible, open-source tool following the OpenScholar model, with a focus on accessibility for under-resourced institutions.
- Integration with publisher workflows. Developing APIs that allow journal submission systems to invoke CitationTrace verification as part of the peer review process, providing editors with automated citation audits.
- Longitudinal provenance tracking. Building a provenance graph that tracks how ideas propagate across papers, enabling not just citation verification but attribution of intellectual lineage , addressing the deeper layers of the provenance problem identified by Earp et al. (2025).
- Adversarial robustness. Evaluating and hardening the system against adversarial inputs, following the robustness dimension of Zhou et al.'s (2024) trustworthy RAG framework.

The provenance problem is one of the most consequential challenges facing AI-assisted scholarship. As LLMs become more deeply embedded in research workflows, the gap between knowledge production and proper attribution will only widen , unless systems like CitationTrace make verification the default, not the exception. By combining retrieval-augmented generation, semantic citation verification, and end-to-end audit logging, CitationTrace demonstrates that it is technically feasible to preserve the chain of scholarly credit in an age of AI-generated knowledge.

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